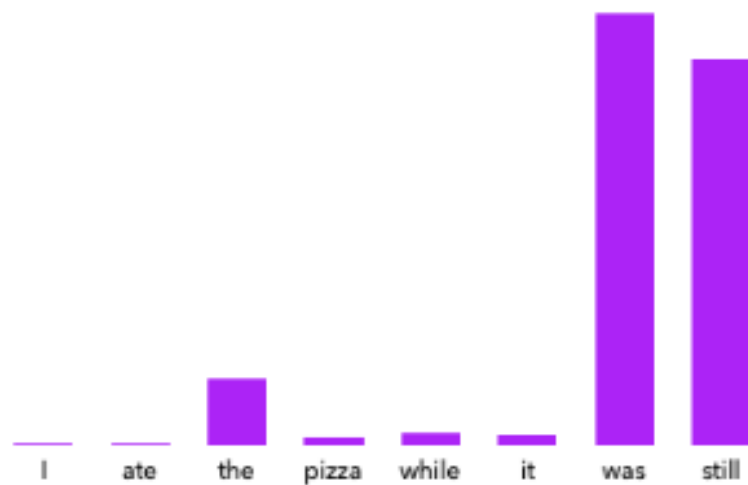


Perceiving Meaning from Sound: Investigating Speech Processing Load with a Conversational Context Task

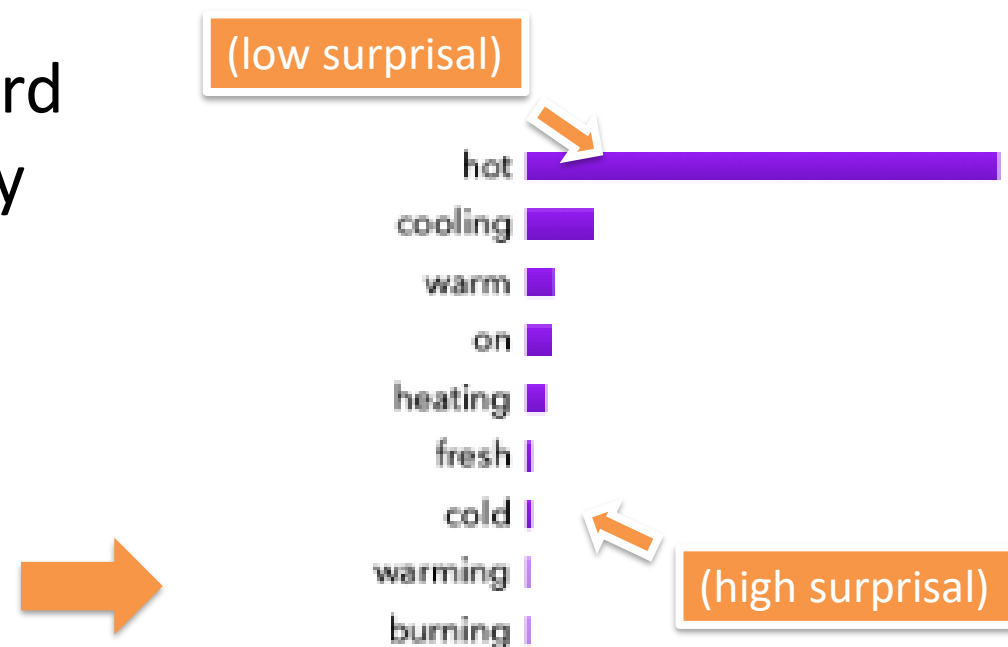
Presenter: Chang D. Liu
Human Verbal Communication Lab
University Tübingen
April 9, 2025

Background: Surprisal

- Surprisal: the unexpectedness of a word given previous words (Hale, 2001; Levy 2008)



I ate the pizza while it was still ...



$$S(w_t) = -\log P(w_t|w_{1:t-1})$$

was
going
n't

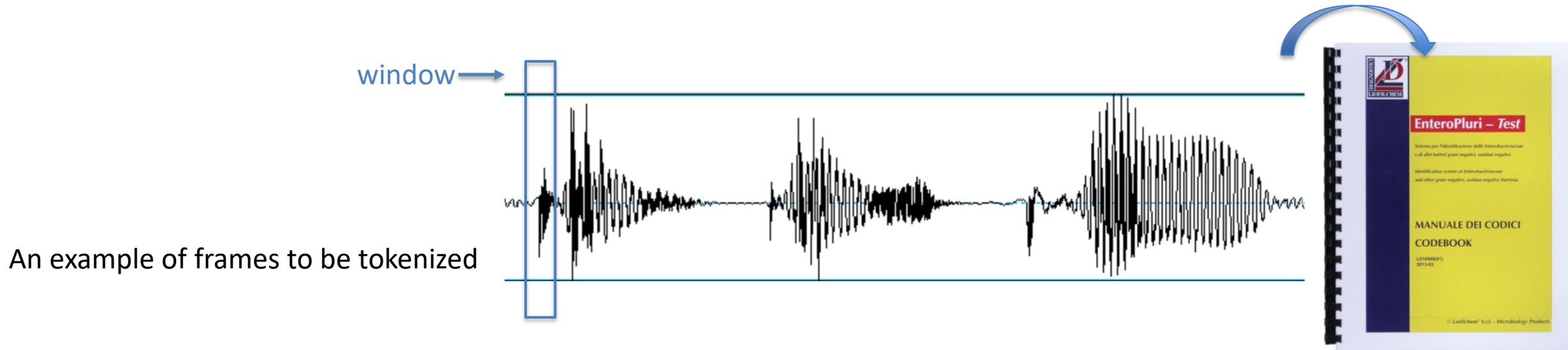
Aim of the study

- Previous studies found linear correlation between reading time and surprisal from LLMs (e.g. Wilcox et al., 2020).
- But LLMs mainly learn dependencies between orthographic words...
- Spontaneous speech, in contrast, contains rich acoustic information that is not captured by the text/word.
- The role of sublexical phonetic variations in speech comprehension remains underexplored.
- Recent advances in speech technology provides the tool to **quantify the uncertainty in speech signals**.

Language modeling in speech generation

- Speech signal → discrete tokens → autoregressive model

Match the frame with the most similar codebook token



Predict the next codebook token

Inference in transformer-based TTS models

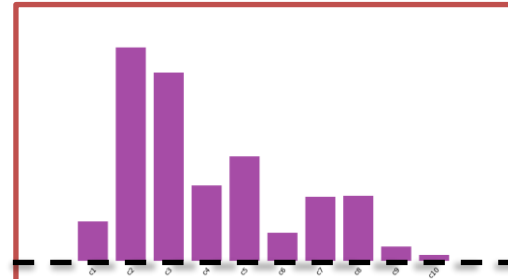
Input tokens

Forward path

Probability of the next codebook token

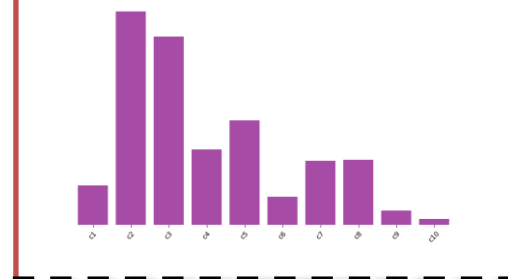
Sampled next token

$t_1, \dots, t_n$



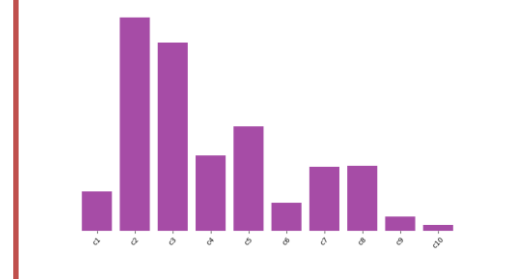
c_1

$t_1, \dots, t_n, c_1$



c_2

$t_1, \dots, t_n, c_1, c_2$

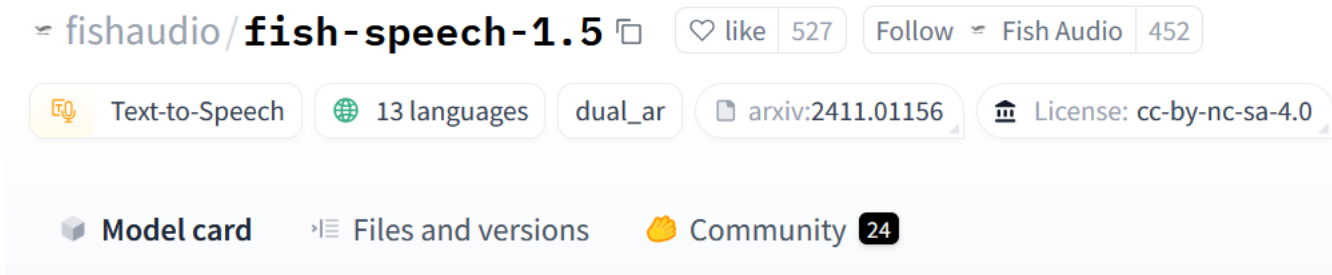


c_3

This can estimate the unexpectedness of the audio

Fish-Speech 1.5

- State-of-the-art open-source TTS model
- Support 13 languages, including German



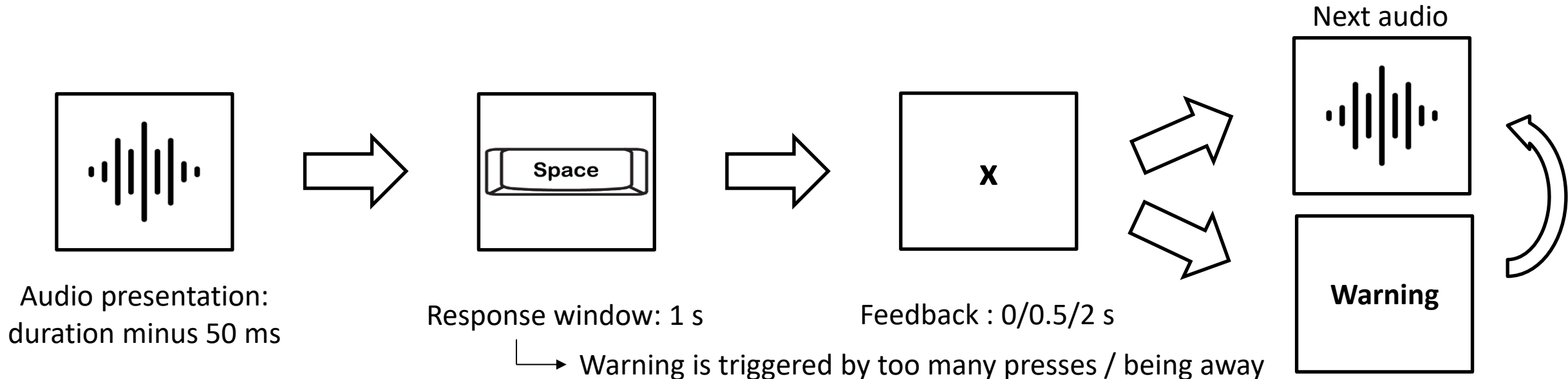
Fish Speech V1.5

Fish Speech V1.5 is a leading text-to-speech (TTS) model trained on more than 1 million hours of audio data in multiple languages.

THE CONVERSATIONAL CONTEXT TASK

The conversational context task

- Material: spontaneous German conversations from the Karl Eberhards Corpus (KEC, Arnold and Tomaschek, 2016)
- Go trials: consecutive phrases from the chosen conversation.
- No-go trials: recordings of the same speaker from distant parts of the conversation that has changed the topic, inserted into random positions



Data analysis

- Linear mixed-effects model: log-transformed response time (logRT)
- Fixed effects: average surprisal of the text tokens (text)

$$-\frac{1}{n} \sum_{i=1}^n \log(P(t_i | t_{i-1}, \dots, t_1))$$

surprisal of the codebook tokens (audio)

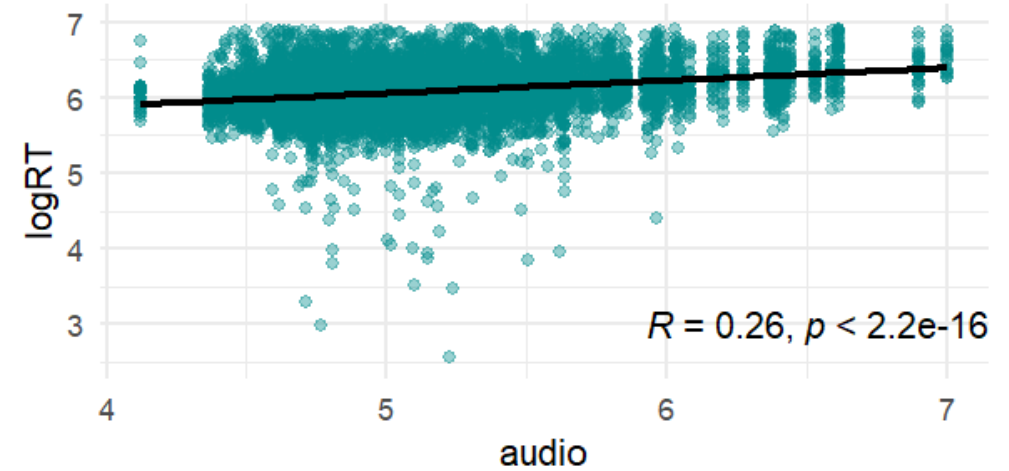
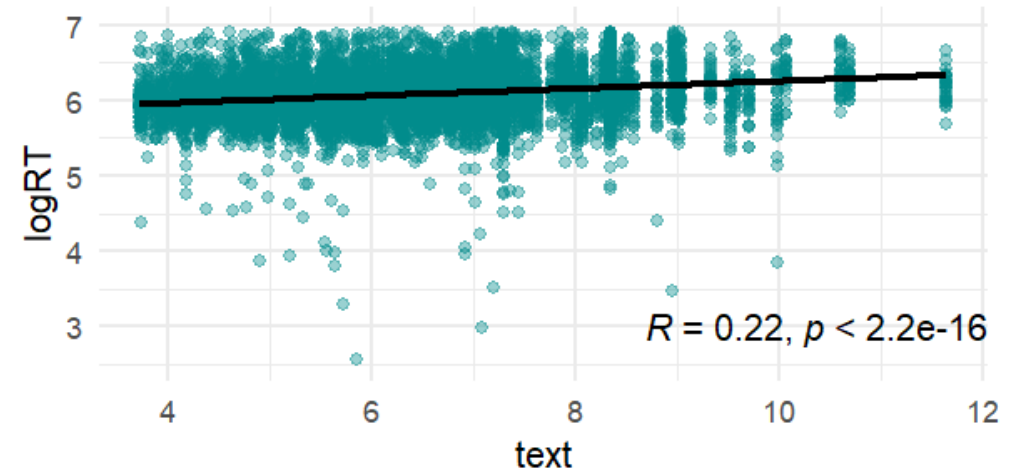
$$-\frac{1}{m} \sum_{j=1}^m \log(P(c_j | c_{j-1}, \dots, c_1, t_n, \dots, t_1))$$

Hypothesis: text and audio surprisal linearly correlates with logRT, and text surprisal should not explain significantly more variance in presence

- Random effects: conversation and participant

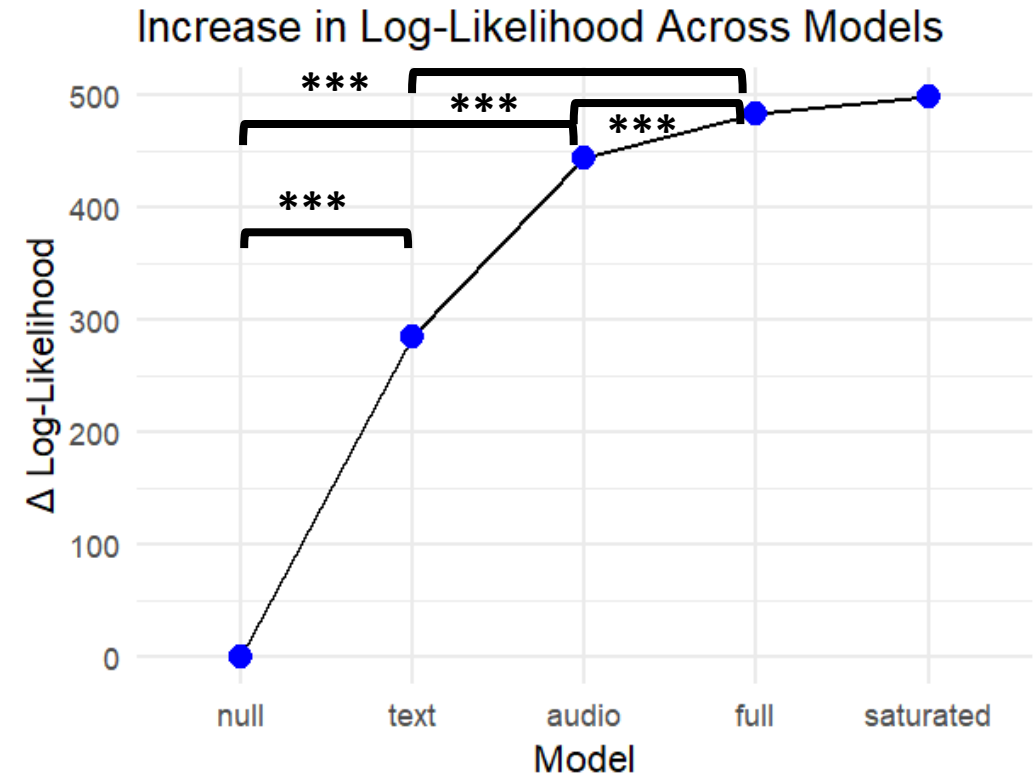
Results

- Linear correlations between surprisal and logRT, replicating previous findings.



Results

- Linear correlations between surprisal and logRT, replicating previous findings.
- LR test: text and audio surprisal significantly improve the null model with intercept only
- LR test: full model with text+audio improves both text and audio model

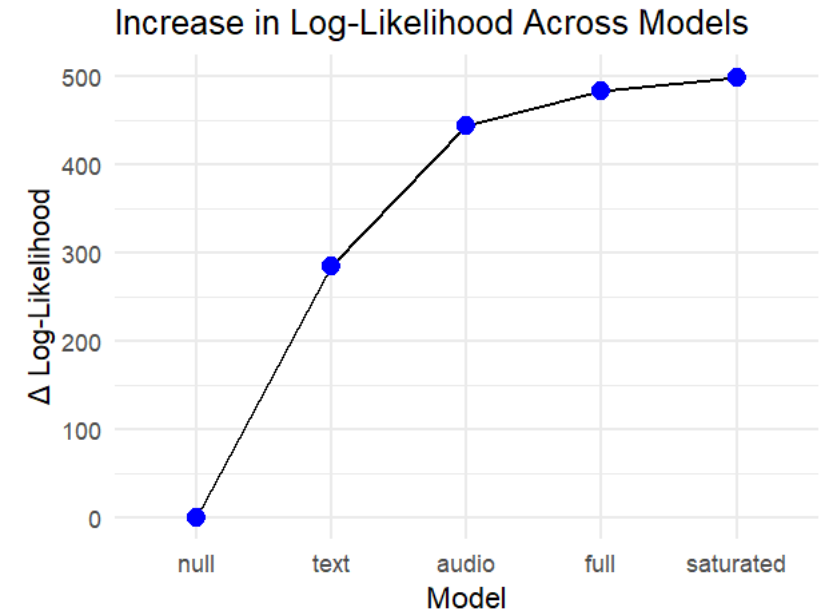


Discussion

- Audio surprisal explains more variance than text surprisal, suggesting that sublexical acoustic cues have a substantial contribution in expectation-based sentence processing.
- Limitation: This method only derive the local phonetic uncertainty without including previous text/audio as context.
- Limitation: duration is not controlled.
- Next step:
 - Include more context
 - Combine measures like pupil dilations and EEG, MEG and iEEG

Conclusion

- The CC task is a feasible behavioral paradigm for people who are interested in the intersection between sentence and speech processing
- The surprisal of the audio plays a larger role than that of the text in speech comprehension.



Q&A

THANK YOU!

My special thanks go to my thesis advisor Konstantin Sering and Michael Franke

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